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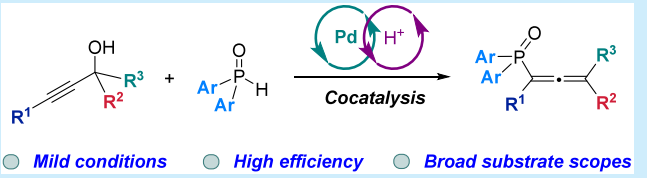
Pd/Brønsted Acid Co-catalyzed Dehydrative Coupling of Propargylic Alcohols with Diarylphosphine Oxides

Zhaoqiang Chen and Hui Qian*

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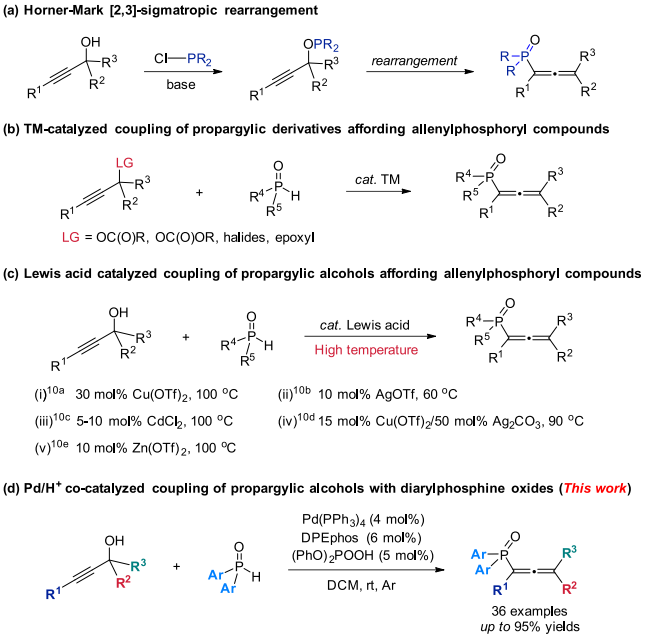
ABSTRACT: An efficient dehydrative coupling of propargylic alcohols with diarylphosphine oxides to construct tetrasubstituted allenylphosphoryl compounds in the presence of a Pd/Brønsted acid co-catalyst has been developed. As a benefit from the use of a Brønsted acid, this reaction could perform under mild conditions with excellent yields, accommodating a wide range of functional groups. The potential utility of this method has also been demonstrated.



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The ongoing advancement of phosphorus chemistry remains a topic of enduring significance due to the diverse applications in pharmaceutical science,[1] materials chemistry,[2] and organic chemistry.[3] Introducing phosphorus atoms into organic molecules could result in substantial modifications to their chemical and physical properties, rendering phosphination a commonly employed efficient method for generating innovative compounds. Allenyl phosphine oxides, specific types of P-functionalized unsaturated compounds, have garnered considerable interest in recent years due to their distinctive allenic structure and phosphorus functionality, which embedded them with unique reactive and biological activities.[4−7] Therefore, increasing attention has been focusing on the advancement of novel methodologies to acquire phosphorus-substituted allenes. The traditional approach for their preparation mainly relies on the Horner−Mark rearrangement reaction, which underwent an intermediate of propargyl phosphate (Scheme 1a).[8] Although delivering the desired allenylphosphoryl compounds, it suffered from toxic and unstable reagents as well as a limited substrate scope. To overcome these issues, transition-metal (TM)-catalyzed coupling of propargylic derivatives with readily available and stable secondary phosphine oxides has been called in for the synthesis of such compounds (Scheme 1b).[9] The direct dehydrate C−P coupling of propargylic alcohols is the most straightforward and efficient strategy; however, the C−O bond is much more inert and less reactive. Therefore, Lewis acid catalysts had been applied in such transformations with a high temperature (Scheme 1c).[10] Inspired by previous work,[11] a catalytic amount of Brønsted acid could facilitate the hydroxy group to form a transient leaving group and could be further converted to construct the C−C bond in the presence of a palladium catalyst. Here, we reported the dehydrative C−P coupling of propargylic alcohols with Pd/H$^+$ co-catalysis, affording the tetrasubstituted allenylphosphoryl compounds under mild conditions with broad functional group compatibility.

Scheme 1. TM-Catalyzed C−P Coupling to Synthesize Allenylphosphoryl Compounds



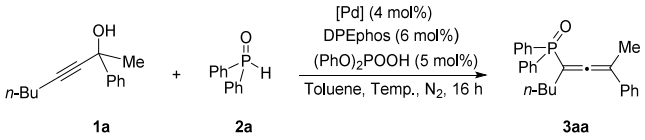
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Table 1. Optimization of the Reaction Conditions[a]



1a 2a 3aa

entry	[Pd]	temperature (°C)	solvent	NMR yield of product **3aa** (%)	recovery of compound **1a** (%)
1	$Pd(PPh_3)_4$	80	toluene	50	30
2[b]	$Pd_2(dba)_3{\cdot}CHCl_3$	80	toluene	45	39
3[b]	$Pd_2(dba)_3$	80	toluene	38	47
4	$Pd(P^tBu_2Ph)_2$	80	toluene	36	46
5[b]	$[(allyl)PdCl]_2$	80	toluene	0	94
6	$PdCl_2$	80	toluene	0	93
7	$Pd(PPh_3)_4$	50	toluene	54	31
8	$Pd(PPh_3)_4$	rt	toluene	52	41
9	$Pd(PPh_3)_4$	rt	MeCN	80	20
10	$Pd(PPh_3)_4$	rt	MTBE	24	76
11	$Pd(PPh_3)_4$	rt	dioxane	38	60
12	$Pd(PPh_3)_4$	rt	THF	40	57
13	$Pd(PPh_3)_4$	rt	DCM	92	0
14[c]	$Pd(PPh_3)_4$	rt	DCM	8	91
15[d]		rt	DCM	0	>99
16[d]	$Pd(PPh_3)_4$	rt	DCM	40	60

[a]Reaction conditions: The reaction was conducted with compound **1a** (0.2 mmol), compound **2a** (0.3 mmol), [Pd] (4 mol %), DPEphos (6 mol %), and $(PhO)_2POOH$ (5 mol %) in solvent (2.0 mL) under a N_2 atmosphere for 16 h. The yield was determined by ^1H NMR analysis of the crude product. [b]The reaction was conducted with 2 mol % [Pd]. [c]The reaction was conducted without $(PhO)_2POOH$. [d]The reaction was conducted without DPEphos.

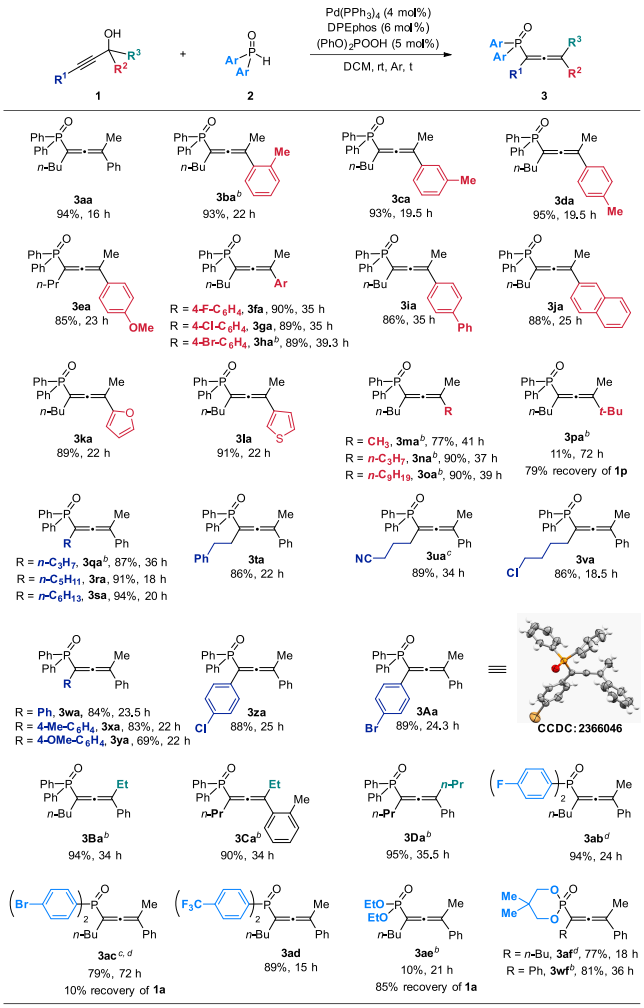
Our research commenced with propargylic alcohol **1a** and diphenyl phosphine oxide **2a** as model substrates to identify the suitable palladium source. When $Pd(PPh_3)_4$ was employed in conjunction with DPEphos and $(PhO)_2POOH$ in toluene, the desired tetrasubstituted allenylphosphoryl product **3aa** was formed in 50% yield with 30% recovery of compound **1a** (entry 1 in Table 1). Other Pd(0) catalysts, such as $Pd_2(dba)_3{\cdot}CHCl_3$, $Pd_2(dba)_3$, and $Pd(P^tBu_2Ph)_2$, could also afford the desired product with lower yields (entries 2−4 in Table 1); however, all of the tested Pd(II) catalysts failed to catalyze the coupling reaction (entries 5 and 6 in Table 1). Further screening indicated that room temperature gave a better result (entries 7 and 8 in Table 1). Next, we set out to investigate the solvent effect. The yield increased to 80% when MeCN was applied (entry 9 in Table 1). Ethers, such as methyl *tert*-butyl ether (MTBE), tetrahydrofuran (THF), and dioxane, were not effective (entries 10−12 in Table 1). Dichloromethane (DCM) was found to be optimal and provided a 92% nuclear magnetic resonance (NMR) yield of product **3aa** (entry 13 in Table 1). Controlling experiments showed that phosphoric acid and palladium are necessary for the reaction (entries 14 and 15 in Table 1). The reaction afforded a 40% yield only in the absence of DPEphos (entry 16 in Table 1). Thus, the optimized reaction conditions entailed the use of 4 mol % $Pd(PPh_3)_4$, 6 mol % DPEphos, and 5 mol % $(PhO)_2POOH$ in DCM at room temperature for 16 h.

With the optimized conditions in hand, we next explored the reaction scope (Scheme 2). Propargylic alcohols bearing a wide range of aryl groups with electron-donating (**1b**−**1e**) and electron-withdrawing (**1f**−**1i**) substituents all reacted smoothly with diphenylphosphine oxide **2a**, forming the corresponding tetrasubstituted allenylphosphoryl products (**3ba**−**3ia**) in decent to excellent yields. It is worth mentioning that the steric effect seems to have no impact because substrates containing a methyl group at the *ortho*, *meta*, and *para* positions of the phenyl group all produced the intended products (**3ba**−**3da**) in significant yields. The reaction of 2-(naphthalen-2-yl)oct-3-yn-2-ol also performed well under the standard conditions, giving an 88% yield of allene product **3ja**. Heteroaryl groups, such as furyl and thienyl, also furnished target products **3ka** and **3la** in decent yields. The R^2 group may also be alkyl groups (**3ma**−**3oa**). However, when a sterically hindered *tert*-butyl group was employed, the allene product **3pa** could be afforded in only a 11% yield with a 79% recovery of raw materials. Then, we set out to investigate the R^1 group with diverse alkyl and aryl substituents: an array of propargylic alcohols bearing different lengths of alkyl chains or containing useful handles were all examined, affording the allene products **3qa**−**3va** in excellent yields. Besides, the R^1 group may also be an aryl group (**1w**−**1A**), providing the related allene products (**3wa**−**3Aa**) and showing a better conversion efficiency with more electron-deficient substrates. The structure of product **3Aa** was confirmed by single-crystal X-ray diffraction analysis. In addition, there was no effect on the reaction when we replaced R^3 with an ethyl group (**1B** and **1C**) or propyl group (**1D**). Meanwhile, reactions of other diarylphosphine oxides, including halides (F and Br) and trifluoromethyl substituents, had also been studied, providing allene possessing different kinds of phosphoryl groups with good yields (**3ab**−**3ad**). Finally, we also performed the C−P coupling reaction with *H*-phosphonates. Diethyl phosphite **2e** exhibited a low reactivity, and only 10% yield of product **3ae** could be supplied. When compound **2e** was replaced with cyclic *H*-phosphonate **2f**, the corresponding products **3af** and **3wf** could be obtained with satisfactory results.

To demonstrate the synthetic potential of the reaction, a gram-scale reaction was conducted to afford 1.4515 g (94% yield) of product **3aa** (eq 1 in Scheme 3). When compound

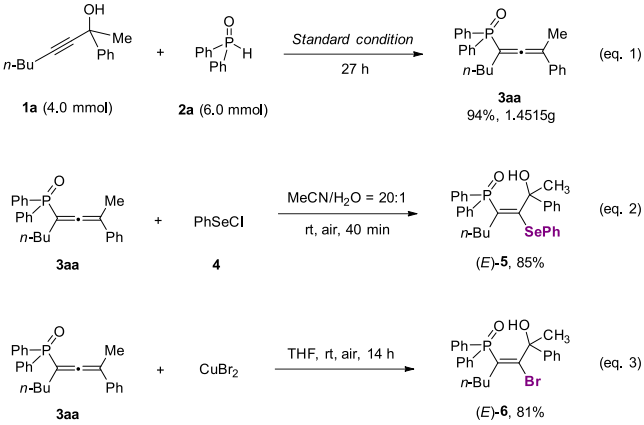
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Scheme 2. Substrate Scope Study[a]



[a]Reaction conditions: The reaction was conducted with compound **1** (1.0 mmol), compound **2** (1.5 mmol), $Pd(PPh_3)_4$ (4 mol %), DPEphos (6 mol %), and $(PhO)_2POOH$ (5 mol %) in DCM (2.5 mL) under an Ar atmosphere. [b]The reaction was conducted with $Pd(PPh_3)_4$ (6 mol %), DPEphos (9 mol %), and $(PhO)_2POOH$ (10 mol %). [c]The reaction was conducted with 1.2 mmol of compound **2**. [d]The reaction was conducted with 10 mol % $(PhO)_2POOH$.

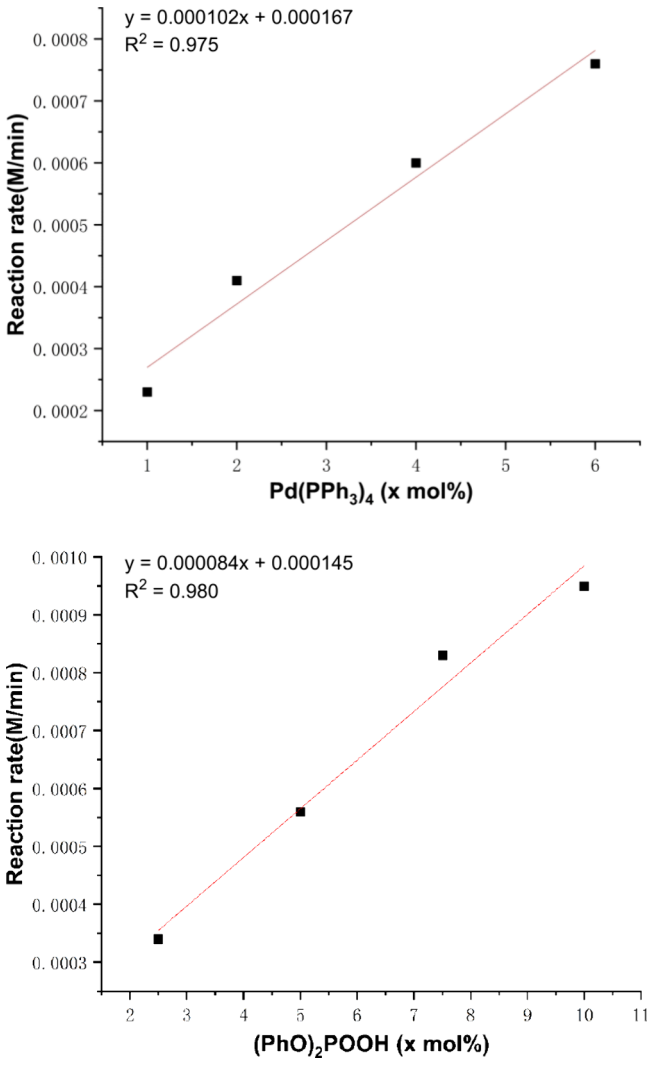
Scheme 3. Gram-Scale Reaction and Transformations of the Products



3aa was treated with an electrophilic reagent, such as phenylselenenyl chloride **4**, the desired addition product (*E*)-**5** could be obtained in 85% yield (eq 2 in Scheme 3).[4b] This allenyl phosphine oxide could also undergo a bromohydroxylation process furnished difunctional compound (*E*)-**6** with 81% yield in the presence of $CuBr_2$ (eq 3 in Scheme 3).[4e]

To gain insight into the mechanism, kinetic studies were conducted and showed that the reaction was first-order in the catalyst and Brønsted acid (Scheme 4). These data indicated

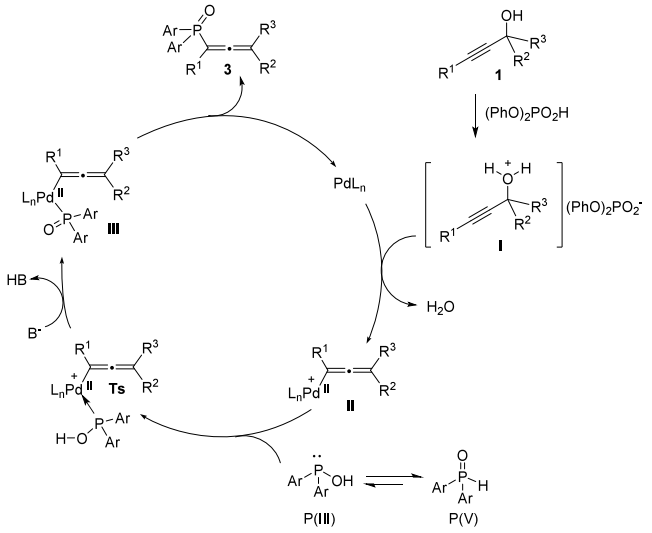
Scheme 4. Kinetic Studies



that both palladium catalyst and phosphoric acid were essential for facilitating the conversion of propargylic alcohol to allene. These findings were also consistent with the results of the controlling experiments (entries 14 and 15 of Table 1). On the basis of the above experimental results and previous reports,[11,12] a possible catalytic cycle was proposed in Scheme 5,[13] in which propargylic alcohol **1** would be activated by Brønsted acid to yield oxonium intermediate **I**, which would easily undergo an *anti*-oxidative addition with palladium(0) to deliver allenyl palladium intermediate **II**. Then, a ligand-exchange process would happen between intermediate **II** and *in situ*-generated tautomeric P(III). Subsequent reductive elimination formed the desired product and regenerated the palladium(0) species.

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Scheme 5. Proposed Mechanism



In summary, we have developed an efficient strategy for the dehydrative coupling to produce the corresponding tetrasubstituted allenylphosphoryl products. The key to our protocol mainly relies on the introduction of Brønsted acid to enhance the leaving ability of the hydroxyl group, which makes the reaction work well under mild conditions with better substrate compatibility. Further studies are currently ongoing in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.4c04586.

Experimental procedures, characterization for obtained compounds, crystal data for compound **3Aa**, and NMR spectra (PDF)

Accession Codes

Deposition number 2366046 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

■ AUTHOR INFORMATION

Corresponding Author

Hui Qian − *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, People's Republic of China;* orcid.org/0000-0002-5606-9213; Email: qian_hui@fudan.edu.cn

Author

Zhaoqiang Chen − *Research Center for Molecular Recognition and Synthesis, Department of Chemistry, Fudan University, Shanghai 200433, People's Republic of China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.4c04586

Notes

The authors declare no competing financial interest.

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